

A Source-Iteration Framework for Monte Carlo Helmholtz Solvers via Walk-on-Boundary



Pratham Gupta¹ Prashanth Shyamala² Benjamin Attal²
Jorge Garcia-Pueyo³ David Lindell²

¹Indian Institute of Science ²University of Toronto ³University of Zaragoza

The problem

We solve the inhomogeneous Helmholtz wave equation $(\Delta + k^2)u = -f$, the PDE governing time-harmonic waves — sound, light, seismic.

FEM/FDM discretize the entire domain. Holding accuracy as $k \rightarrow \infty$ needs DOFs growing faster than k^d (the pollution effect) — slow, memory-heavy, and the indefinite linear system is hostile to solve. Adjoint/FWI-style inversion must additionally store the full wavefield, $O(k^d)$ per frequency, and runs out of memory at scale.

Monte Carlo has no global dependence. Grid-free, output-sensitive estimators — Walk-on-Spheres/Walk-on-Boundary [1] — evaluate u at any query point directly, with no mesh and no global linear system to assemble or store.

But existing MC methods fail for Helmholtz. The boundary operator A is sign-indefinite, so $\rho(A) \geq 1$ and the Neumann series $\mu = \sum_{n \geq 0} A^n b$ diverges — the walk never converges.

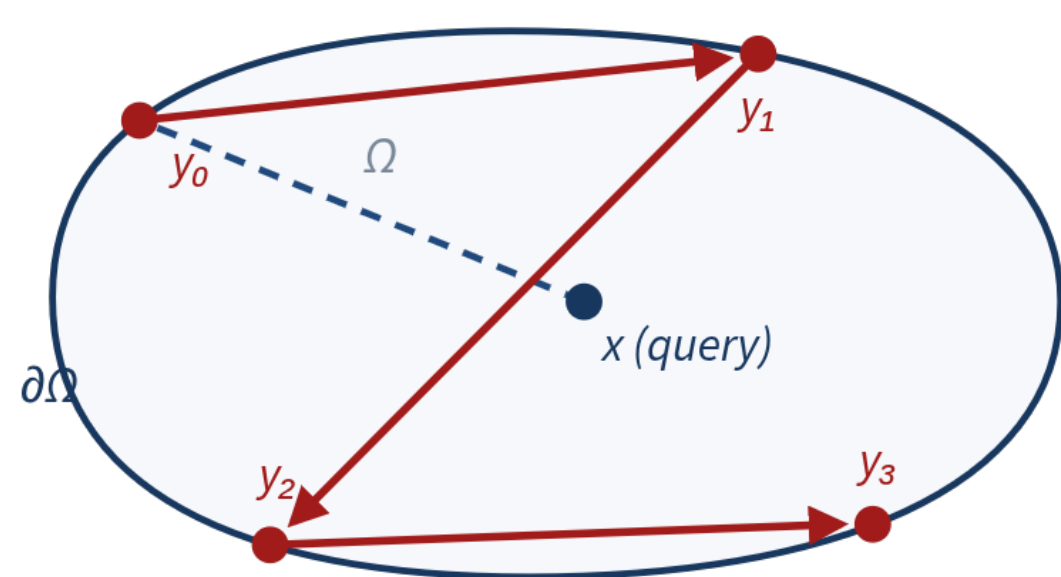
Walk on Boundary

Walk-on-Boundary recasts the BVP as an integral equation along $\partial\Omega$,

$$\mu(x) - \int_{\partial\Omega} K(x, y) \mu(y) dS_y = b(x), \quad x \in \partial\Omega$$

(compactly $(I - A)\mu = b$),

and estimates the density μ with unbiased random walks on $\partial\Omega$ that realize the Neumann series $\mu = \sum_{n \geq 0} A^n b$, without ever discretizing the interior of Ω .



Why it breaks for Helmholtz

$$(I - A)\mu = b, \quad \mu = \sum_{n \geq 0} A^n b \quad \text{converges only if } \rho(A) < 1$$

At moderate to high k the indefinite kernel gives $\rho(A) \geq 1$: the series **diverges**, and deep walks estimating A^n have unbounded variance — the failure the Convergent Born Series [3] first exposed.

Related Work

Grid-free MC solvers. WoS for (screened) Poisson [1]; Walk-on-Stars adds Neumann and Robin data.

Walk-on-Boundary. Practical boundary-integral estimator for Dirichlet / Neumann / Robin data [2]; convergence guarantees are missing for indefinite operators.

Convergent Born Series. Complex wavenumber $k_0^2 + i\varepsilon$ and $\gamma = (i/\varepsilon)V$ force $\rho(M) < 1$ unconditionally [3] — the deterministic precedent for my shift.

Closest precedent. WoS for Helmholtz / Yukawa via the Duffin correspondence [4] — but tied to constant potentials.

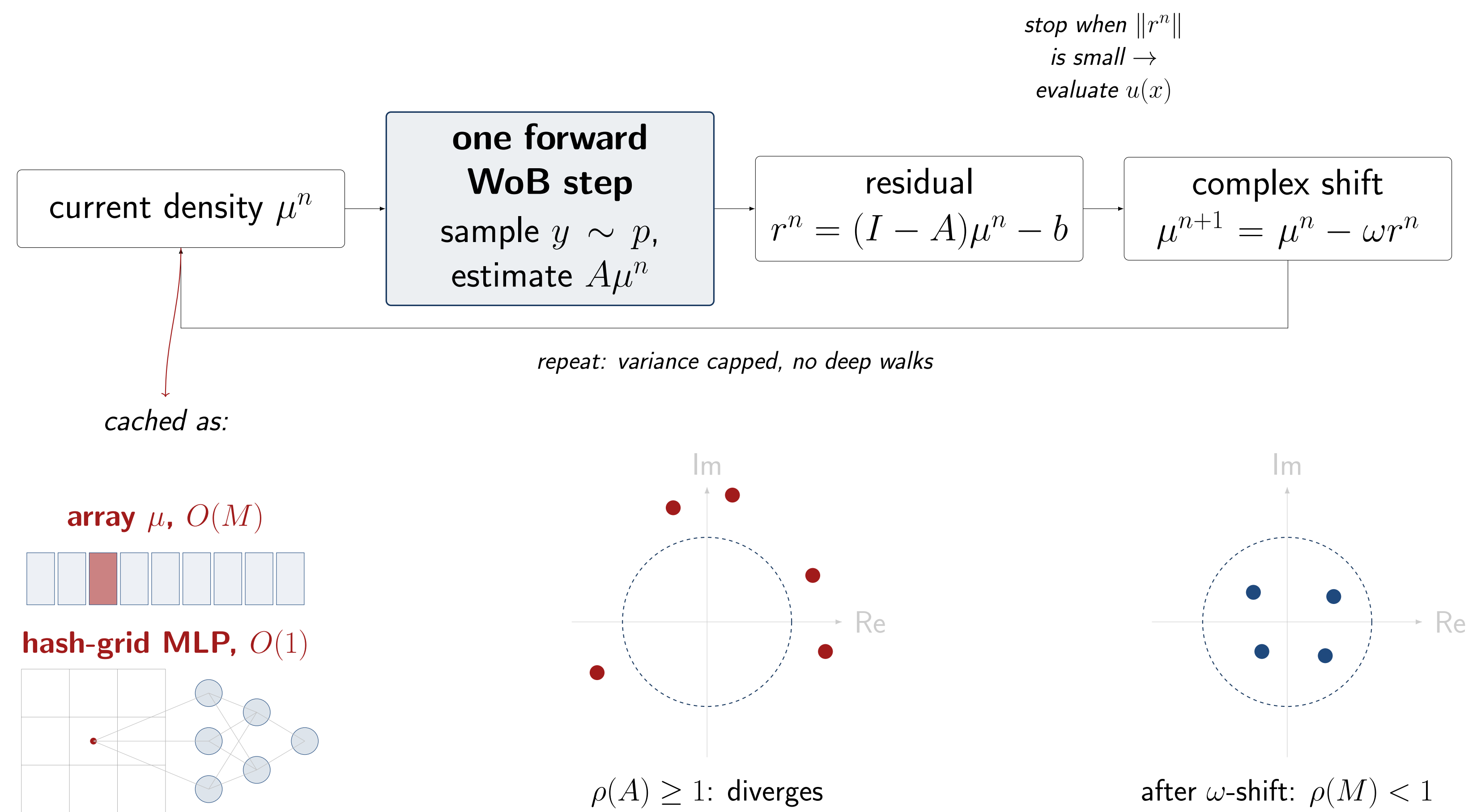
References

- R. Sawhney and K. Crane. "Monte Carlo Geometry Processing: A Grid-Free Approach to PDE-Based Methods on Volumetric Domains". In: *ACM Transactions on Graphics* 39.4 (2020).
- R. Sugimoto, T. Chen, Y. Jiang, C. Batty, and T. Hachisuka. "A Practical Walk-on-Boundary Method for Boundary Value Problems". In: *ACM Transactions on Graphics* 42.4 (2023).
- G. Osnabrugge, S. Leedumrongwatthanakun, and I. M. Vellekoop. "A Convergent Born Series for the Inhomogeneous Helmholtz Equation in Arbitrarily Large Media". In: *Journal of Computational Physics* 322 (2016), pp. 113–124.
- X. Yang, A. Rasila, and T. Sottinen. "Walk on Spheres Algorithm for Helmholtz and Yukawa Equations via Duffin Correspondence". In: *Methodology and Computing in Applied Probability* 19.2 (2017), pp. 589–602.
- W. Jakob, S. Speierer, N. Roussel, and D. Vicini. "Dr.Jit: A Just-In-Time Compiler for Differentiable Rendering". In: *ACM Transactions on Graphics* 41.4 (2022).

Methodology

The source iteration, one turn of the loop

one iteration = one operator apply



$$\begin{aligned} \mu^{n+1} &= \mu^n - \omega [(I - A)\mu^n - b] \\ M &= I - \omega(I - A), \quad \rho(M) < 1 \\ \widehat{A}\mu^n &= \frac{K(x, y) \mu^n(y)}{p(y)}, \quad y \sim p \end{aligned}$$

A complex ω rotates and rescales the spectrum of $I - A$ into the unit disk about 1 — the stochastic analog of the CBS ε -shift [3].

One boundary sample per apply: unbiased, matrix-free, variance independent of n .

No deep walks. Every operator apply is one forward WoB step, so per-iteration variance is capped — the exponential blow-up of estimating A^n disappears.

Warm starts. Bootstrap μ^0 from a geometrical-optics / Kirchhoff approximation, a neighbouring frequency, or a learned cache $\hat{\mu}_\theta$ trained by a semi-gradient residual loss.

Implementation. Matrix-free applies and forward walks on Dr.Jit [5] with Mitsuba-style ray tracing: JIT-compiled, GPU/CPU-retargetable, differentiable.

Evaluation

What I will trust as evidence — figures and table are placeholders, not measurements.

Figure 1 — residual vs. iteration (placeholder)

Figure 1. $\log \|(I - A)\mu^n - b\|$ vs. iteration n , source iteration against the Neumann series over $\rho(A) \in \{0.5, \dots, 1.5\}$. Falsifiable prediction: Neumann curves turn upward at $\rho \geq 1$; source iteration descends at every ρ .

Figure 2 — $\rho(M)$ over the complex ω plane (placeholder)

Figure 2. Measured decay rate $\rho(M)$ as a function of $\{\text{Re } \omega, \text{Im } \omega\}$. Prediction: a convergent basin bounded by the contour $\rho(M) = 1$, with an optimum ω^* inside it.

Method	$\rho = 0.7$	$\rho = 0.95$	$\rho = 1.05$	$\rho = 1.3$
Neumann-series WoB	converges (slow)	converges (very slow)	diverges	diverges
Source iteration (tuned ω)	converges	converges	converges	converges

Table 1. (Illustrative placeholder.) Outcome vs. spectral-radius sweep — the crossover at $\rho = 1$ is the test.

Reference solutions. Analytic fields where they exist, otherwise a dense deterministic boundary-element solve — error reported against both, never against a single baseline.

Controls. Sweep $\rho(A)$ across the $\rho = 1$ boundary; report per-iteration variance and sample counts, so convergence claims cannot hide behind extra samples.

What it means

Convergence becomes a design choice, not a property of the domain. One complex scalar ω buys $\rho(M) < 1$ where the Neumann series has none.

Robustness is paid for in rate. Following the CBS law $\Delta r = 2k_0/\varepsilon$, a larger shift guarantees convergence but slows each iteration — warm starts buy the rate back near $\rho \approx 1$.

Open questions I want to discuss. How far does a scalar ω stretch before an operator-valued shift is needed for heterogeneous media (Lippmann–Schwinger), and can $\hat{\mu}_\theta$ warm-start inverse problems without biasing gradients?